

Automated Retail Analytics Pipeline

□ Project Overview

This project simulates an end-to-end automated analytics pipeline for a retail organization.

The system extracts transactional sales data, performs data cleaning and transformation (ETL), stores structured data in a SQLite database, and generates automated KPI reports.

The objective is to demonstrate real-world data engineering and analytics workflow using Python and SQLite.

□ Business Problem

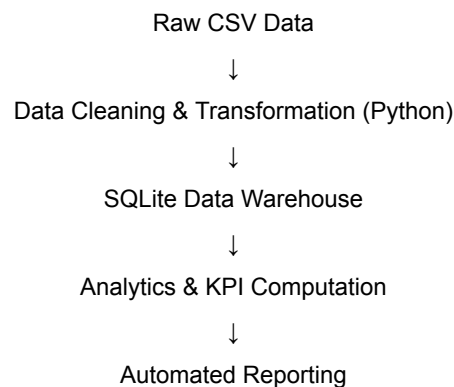
Retail managers manually generate daily and monthly sales reports from CSV files. This process is:

- Time-consuming
- Error-prone
- Not scalable

The company requires an automated system to:

- Clean raw sales data
- Store it centrally
- Generate KPI metrics
- Support faster decision-making

□ System Architecture



□ Data Source

Dataset: Online Retail Dataset

Type: Transaction-level sales data

Key Columns:

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

□ ETL Process

Extract

- Load CSV file using pandas

Transform

- Remove duplicate rows
- Remove cancelled orders (negative quantities)
- Handle missing CustomerID
- Create new column: $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$
- Convert InvoiceDate to datetime format

Load

- Store cleaned data into SQLite database

□ Storage Design

SQLite was selected because:

- Lightweight and serverless
- Suitable for academic projects
- Easy integration with Python

The database follows a simplified star-schema design.

Scheduling

The ETL process is designed to run automatically using scheduled execution (simulated cron-based scheduling).

Future improvement includes integration with Apache Airflow.

Logging & Monitoring

- Python logging module is used to track pipeline execution.
- Logs include:
 - Start time
 - End time
 - Number of rows processed
 - Error handling

Reporting Layer

The system generates:

- Daily revenue
- Monthly revenue
- Top 10 products
- Sales by country
- Repeat customer analysis

Tools & Technologies

- Python
- Pandas
- SQLite
- Jupyter Notebook
- Git & GitHub

How to Run

1. Clone repository
2. Create virtual environment
3. Install dependencies
4. Run ETL script
5. Launch analytics notebook

□ Future Improvements

- Integration with Apache Airflow
- Deployment on cloud
- Real-time streaming data
- Interactive dashboard using Streamlit

step to run the project

- `git clone cd Retail_Project`
- `python -m venv venv` for mac and `venv\Scripts\activate` for windows
- `pip install -r requirements.txt`
- `python etl.py`
- `python rfm.py`
- `streamlit run app.py`